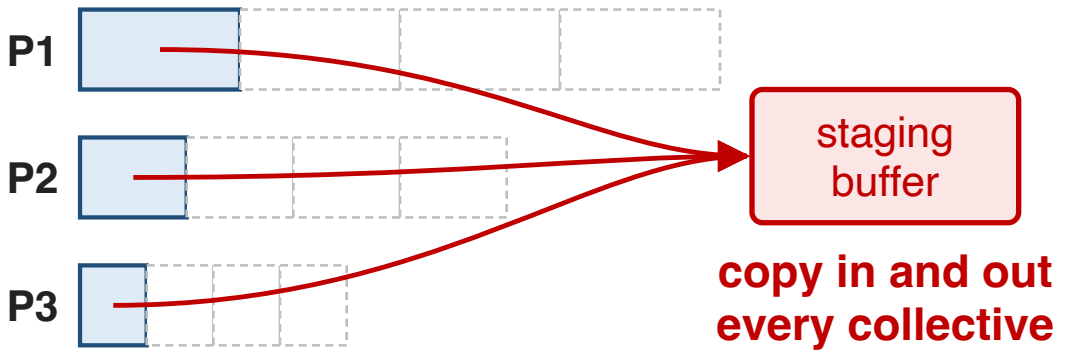
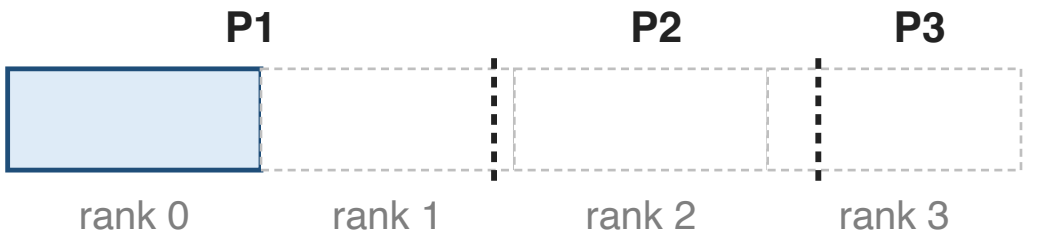
(a) Per-parameter shards (FSDP2)

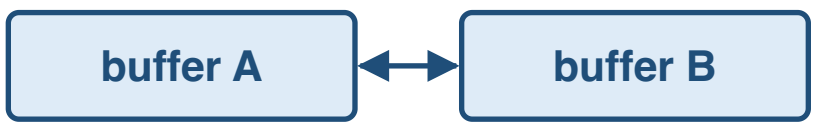


rank 0 owns a quarter of each parameter

(b) Flat per-module buffer



AG and RS run in place on NCCL-registered memory



units alternate between two buffers